

In general, when the phrase "without loss of generality" is used in a proof (often abbreviated as WLOG), we assert that by proving one case of a theorem, no additional argument is required to prove other specified cases. That is, other cases follow by making straightforward changes to the argument, or by filling in some straightforward initial step.

Example 7: Show that if x and y are integers and both xy and $x+y$ are even, then both x and y are even.

Solution: $p: xy$ is even $q: x+y$ is even $r: x$ is even $s: y$ is even.

We shall prove $(p \wedge q) \rightarrow (r \wedge s)$ is True by contradiction.

$\therefore$ We need to show that $(p \wedge q) \wedge \neg(r \wedge s) \rightarrow F$ is True.

Let, $(p \wedge q) \wedge \neg(r \wedge s)$ be True. By defn. of AND, $(p \wedge q)$ is True and $\neg(r \wedge s)$ is True. By defn. of AND, both p and q are respectively True. By De Morgan's Law, $(\neg r \vee \neg s)$ is True.

$\neg r: x$ is not even, i.e. x is odd (By Defn. 1)
 $\neg s: y$ is not even, i.e. y is odd (By Defn. 1)

From the defn. of OR, the following 3 cases arise:

① $\neg r$ is True, $\neg s$ is False. $\therefore \neg(\neg s)$ is True (From defn. of negation)

$\therefore s$ is True (By double-negation Law)

So, x is odd and y is even

② $\neg r$ is False, $\neg s$ is True. $\therefore \neg(\neg r)$ is True (From defn. of negation)

$\therefore r$ is True (By double-negation Law)

So, x is even, y is odd.

Since x and y are two arbitrary integers, we can combine the cases 1 and 2 into a single case. $\therefore$ WLOG, let x is odd and y is even.

By Defn 1, $\exists$ integers K_1 and K_2 s.t. $x = 2K_1 + 1$ and $y = 2K_2$